

non-convex



NP-hard

regularize with entropy

convexifies + enables Sinkhorn-like iterations



for each iteration

Gromov-Wasserstein: Computation

- Objective is **non-convex**, quadratic in $\pi \Rightarrow$ multiple local minima
- Equivalent to a Quadratic Assignment Problem (QAP) $\rightarrow$ **NP-hard**.
- Requires iterative or approximate solvers.

- As always, can **regularize with entropy**:

$$\min_{\pi \in \Pi(\mathbf{a}, \mathbf{b})} E_{\mathbf{D}, \mathbf{D}'}(\pi) - \varepsilon H(\pi), \quad E_{\mathbf{D}, \mathbf{D}'}(P) = \sum_{i, i', j, j'} |\mathbf{D}_{ii'} - \mathbf{D}'_{jj'}|^2 P_{ij} P_{i'j'}$$

- This **convexifies + enables Sinkhorn-like iterations** (Solomon et al. 2016):

- At iteration t :

1. $\mathbf{C}^{(t)} \leftarrow -\mathbf{D} \pi^{(t)} \mathbf{D}'$ ($= \nabla E_{\mathbf{D}, \mathbf{D}'}(\pi^{(t)})$, local linear approximation of the quadratic term)

2. $\pi^{(t+1)} \leftarrow \min_{\pi \in \Pi(\mathbf{a}, \mathbf{b})} \langle \pi, \mathbf{C}^{(t)} \rangle - \varepsilon H(\pi)$ (solved via Sinkhorn)

- Notes:

- solves an entire (regularized) OT problem **for each iteration**
- can be interpreted as a mirror-descent scheme ($\rightarrow$ KL projection of gradient into constraint set)

Gromov-Wasserstein: Application

Unsupervised word translation (AM & Jaakkola, 2018)

